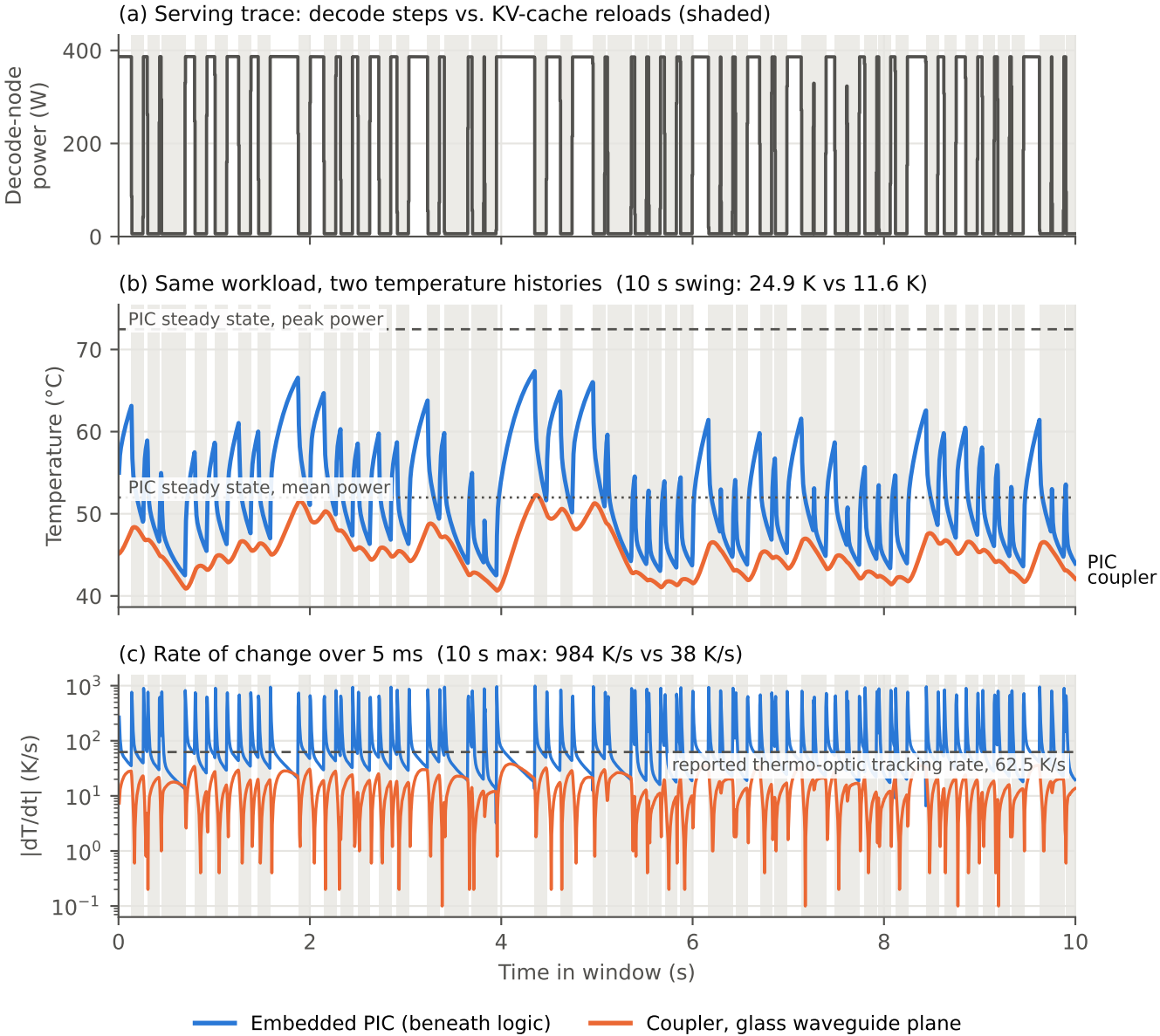
(a) Serving trace: decode steps vs. KV-cache reloads (shaded)
Decode-node power (W)
400
200
0
(b) Same workload, two temperature histories (10 s swing: 24.9 K vs 11.6 K)
PIC steady state, peak power
PIC steady state, mean power
Temperature (°C)
70
60
50
40
PIC
coupler
(c) Rate of change over 5 ms (10 s max: 984 K/s vs 38 K/s)
|dT/dt| (K/s)
$10^3$
$10^2$
$10^1$
$10^0$
$10^{-1}$
reported thermo-optic tracking rate, 62.5 K/s
0
2
4
6
8
10
Time in window (s)
Embedded PIC (beneath logic)
Coupler, glass waveguide plane